# Report_analisi_esplorativa

2026-04-14

```
knitr::opts_chunk$set(
  echo = TRUE,
  warning = FALSE,
  message = FALSE,
  error = FALSE,
  fig.pos = 'H',
  out.width = "100%")

source("Libraries.R")
```

```
## Warning: il pacchetto 'tidyverse' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'tidyr' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'readr' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'purrr' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'dplyr' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'lubridate' è stato creato con R versione 4.3.3
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Warning: il pacchetto 'gtsummary' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'naniar' è stato creato con R versione 4.3.3
## Caricamento del pacchetto richiesto: viridisLite
## Warning: il pacchetto 'ggrepel' è stato creato con R versione 4.3.3
## Warning: il pacchetto 'pals' è stato creato con R versione 4.3.3
##
## Caricamento pacchetto: 'pals'
##
## I seguenti oggetti sono mascherati da 'package:viridis':
##
##     cividis, inferno, magma, plasma, turbo, viridis
##
## I seguenti oggetti sono mascherati da 'package:viridisLite':
```

```
##
##     cividis, inferno, magma, plasma, turbo, viridis
```

```
merged_data <- read.csv("Input/merged_data.csv")

# metto a posto la tipologia di variabile
cols_date <- c("Data_di_nascita", "Data_esordio", "Data_diagnosi", "Data_prelievo_T0", "data_referto_T6
merged_data[cols_date] <- lapply(merged_data[cols_date], as.Date)
cols_factor <- c("lesioni_T6", "relapse_T6", "lesioni_T12", "relapse_T12",
                 "lesioni_T18", "relapse_T18", "lesioni_T24", "relapse_T24",
                 "lesioni_T30", "relapse_T30", "lesioni_T36", "relapse_T36",
                 "cambio_terapia", "lesion_burden", "bone_marrow_lesions",
                 "subtentorial_lesions", "gadolinium_contrast", "full_recovery", "naive", "smoking_habit
merged_data <- merged_data %>%
  mutate(
    across(all_of(cols_factor), ~ case_when(
      .x %in% c("No", "no", "0", 0) ~ "No",
      .x %in% c("Yes", "yes", "1", 1) ~ "Yes",
      TRUE ~ NA_character_
    ) %>% factor(levels = c("No", "Yes"))))

cols_factor2 <- c("glucocorticoid_treatment", "gc_treatment", "prognosis_at_onset")
merged_data[cols_factor2] <- lapply(merged_data[cols_factor2], factor, levels = c("Positive", "Negative
merged_data$physical_activity <- factor(merged_data$physical_activity, levels = c("active", "sedentary"
merged_data$simplified_batch <- factor(merged_data$simplified_batch, levels = c("1-3", "4-5"))
merged_data$disease_activity <- factor(merged_data$disease_activity, levels = c("higher_activity", "low
merged_data$sex <- factor(merged_data$sex, levels = c("M", "F"))
merged_data$sequencing_batch <- factor(merged_data$sequencing_batch, levels = c("1", "2", "3"))
merged_data$CENTRO <- factor(merged_data$CENTRO, levels = c("CLERICO", "NAPOLI", "MOLINETTE", "CRESM"))
merged_data$clinical_presentation <- factor(merged_data$clinical_presentation)

merged_data <- merged_data %>%
  mutate(across(starts_with("EDSS"), ~ {x_char <- trimws(as.character(.x))
    ordered(x_char, levels = as.character(seq(0, 10, by = 0.5)))}))

cols_numeric <- c("MEDAS", "cortisone_T6", "cortisone_T12", "cortisone_T18", "cortisone_T24", "cortison
merged_data[cols_numeric] <- lapply(merged_data[cols_numeric], as.numeric)

#Pulizia x evitare crash LaTeX
merged_data$NOTE <- gsub("[#$%&_]", " ", merged_data$NOTE)
merged_data$nuova_terapia <- gsub("[#$%&_]", " ", merged_data$nuova_terapia)
merged_data$previous_therapy <- gsub("[#$%&_]", " ", merged_data$previous_therapy)
```

```
summary(merged_data)
```

```
##       ID             Data_di_nascita          MEDAS             CENTRO
##  Length:30          Min.   :1966-05-17   Min.   :35.00   CLERICO  :22
##  Class :character   1st Qu.:1976-08-24   1st Qu.:42.20   NAPOLI   : 3
##  Mode  :character   Median :1983-08-26   Median :42.90   MOLINETTE: 1
##                     Mean   :1984-03-12   Mean   :50.55   CRESM    : 4
##                     3rd Qu.:1992-03-26   3rd Qu.:62.50
##                     Max.   :2002-08-29   Max.   :85.70
##                                          NA's   :8
##   Data_esordio        Data_diagnosi        Data_prelievo_T0        EDSS_T0
##  Min.   :2000-08-01   Min.   :2000-08-01   Min.   :2020-01-10   1      :8
```

```
##  1st Qu.:2015-07-01   1st Qu.:2017-12-06   1st Qu.:2021-06-30   2      :6
##  Median :2020-06-01   Median :2021-02-24   Median :2022-01-24   1.5    :4
##  Mean   :2017-10-14   Mean   :2019-01-04   Mean   :2022-01-23   2.5    :4
##  3rd Qu.:2021-04-15   3rd Qu.:2021-10-28   3rd Qu.:2022-09-18   0      :2
##  Max.   :2022-12-12   Max.   :2023-03-30   Max.   :2023-04-04   (Other):5
##  NA's   :1                                                      NA's   :1
##  data_referto_T6         EDSS_T6   lesioni_T6 relapse_T6  cortisone_T6
##  Min.   :2021-06-24   1      : 5   No  :15    No  :14    Min.   : 0.000
##  1st Qu.:2021-12-06   2      : 3   Yes : 2    Yes : 3    1st Qu.: 0.000
##  Median :2022-10-25   0      : 2   NA's:13    NA's:13    Median : 0.000
##  Mean   :2022-10-21   1.5    : 2                         Mean   : 1.294
##  3rd Qu.:2023-09-11   2.5    : 1                         3rd Qu.: 0.000
##  Max.   :2023-11-16   (Other): 2                         Max.   :15.000
##  NA's   :18           NA's   :15                         NA's   :13
##  data_referto_T12        EDSS_T12 lesioni_T12 relapse_T12 cortisone_T12
##  Min.   :2020-12-22   1      :8   No  :21     No  :24     Min.   :0
##  1st Qu.:2022-07-10   0      :5   Yes : 3     Yes : 0     1st Qu.:0
##  Median :2023-01-17   1.5    :4   NA's: 6     NA's: 6     Median :0
##  Mean   :2023-01-16   2      :3                          Mean   :0
##  3rd Qu.:2023-08-23   2.5    :2                          3rd Qu.:0
##  Max.   :2024-05-09   (Other):6                          Max.   :0
##  NA's   :11           NA's   :2                          NA's   :10
##  data_referto_T18        EDSS_T18  lesioni_T18 relapse_T18 cortisone_T18
##  Min.   :2022-01-04   1      : 4   No  :13     No  :15     Min.   :0
##  1st Qu.:2023-03-09   1.5    : 3   Yes : 3     Yes : 1     1st Qu.:0
##  Median :2023-08-29   0      : 2   NA's:14     NA's:14     Median :0
##  Mean   :2023-10-15   2      : 2                         Mean   :0
##  3rd Qu.:2024-08-15   4      : 1                         3rd Qu.:0
##  Max.   :2024-12-23   (Other): 2                         Max.   :0
##  NA's   :16           NA's   :16                         NA's   :14
##  data_referto_T24        EDSS_T24  lesioni_T24 relapse_T24 cortisone_T24
##  Min.   :2022-03-16   1      : 4   No  :14     No  :15     Min.   :0.0000
##  1st Qu.:2023-08-15   0      : 3   Yes : 3     Yes : 2     1st Qu.:0.0000
##  Median :2024-02-19   1.5    : 3   NA's:13     NA's:13     Median :0.0000
##  Mean   :2024-03-23   2      : 3                         Mean   :0.5385
##  3rd Qu.:2025-02-15   2.5    : 2                         3rd Qu.:0.0000
##  Max.   :2025-08-05   (Other): 3                         Max.   :4.0000
##  NA's   :18           NA's   :12                         NA's   :17
##  data_referto_T30        EDSS_T30  lesioni_T30 relapse_T30 cortisone_T30
##  Min.   :2022-10-19   0      : 3   No  : 9     No  : 9     Min.   :0.0000
##  1st Qu.:2024-02-21   1.5    : 3   Yes : 3     Yes : 3     1st Qu.:0.0000
##  Median :2024-05-23   1      : 2   NA's:18     NA's:18     Median :0.0000
##  Mean   :2024-06-04   2      : 2                         Mean   :0.5833
##  3rd Qu.:2024-10-07   4.5    : 1                         3rd Qu.:0.0000
##  Max.   :2025-05-30   (Other): 1                         Max.   :4.0000
##  NA's   :21           NA's   :18                         NA's   :18
##  data_referto_T36        EDSS_T36  lesioni_T36 relapse_T36 cortisone_T36
##  Min.   :2023-04-07   1      : 2   No  : 4     No  : 5     Min.   :0.0
##  1st Qu.:2024-02-15   0      : 1   Yes : 1     Yes : 0     1st Qu.:0.0
##  Median :2025-02-10   1.5    : 1   NA's:25     NA's:25     Median :0.0
##  Mean   :2024-09-30   6.5    : 1                         Mean   :0.6
##  3rd Qu.:2025-07-01   0.5    : 0                         3rd Qu.:0.0
##  Max.   :2025-10-01   (Other): 0                         Max.   :3.0
##  NA's   :25           NA's   :25                         NA's   :25
```

```
##  cambio_terapia data_cambio_terapia  nuova_terapia          NOTE
##  No  :14        Min.   :2022-01-01   Length:30          Length:30
##  Yes : 7        1st Qu.:2023-11-21   Class :character   Class :character
##  NA's: 9        Median :2024-06-16   Mode  :character   Mode  :character
##                 Mean   :2024-03-11
##                 3rd Qu.:2024-12-03
##                 Max.   :2025-07-15
##                 NA's   :22
##    therapy           naive    previous_therapy     sex       age_at_t0
##  Length:30          No  :11   Length:30          M   : 9   Min.   :20.00
##  Class :character   Yes :18   Class :character   F   :20   1st Qu.:30.25
##  Mode  :character   NA's: 1   Mode  :character   NA's: 1   Median :38.50
##                                                           Mean   :37.37
##                                                           3rd Qu.:44.25
##                                                           Max.   :55.00
##
##       bmi        smoking_habit        physical_activity antibiotic_use
##  Min.   :16.80   No  :17       active           :5       No  :16
##  1st Qu.:20.10   Yes : 7       sedentary        :9       Yes : 2
##  Median :21.80   NA's: 6       partially_active:7       NA's:12
##  Mean   :23.54                 NA's             :9
##  3rd Qu.:26.50
##  Max.   :32.10
##  NA's   :9
##  sequencing_batch spike_in  dna_quantification sample_type
##  1   :16          No  :16   Min.   : 62.76     Length:30
##  2   : 0          Yes : 2   1st Qu.: 84.06     Class :character
##  3   : 2          NA's:12   Median :105.36     Mode  :character
##  NA's:12                    Mean   :105.36
##                             3rd Qu.:126.66
##                             Max.   :147.96
##                             NA's   :28
##  glucocorticoid_treatment simplified_batch        disease_activity
##  Positive:18              1-3 :18          higher_activity:18
##  Negative: 0              4-5 : 0          lower_activity : 0
##  NA's    :12              NA's:12          NA's           :12
##
##
##
##
##  sample_collection_date glc_date_before_t0     gc_treatment days_gc_collection
##  Min.   :2020-01-10     Min.   :2019-11-16   Positive:12    Min.   : 38.0
##  1st Qu.:2021-08-10     1st Qu.:2021-04-22   Negative: 6    1st Qu.: 69.0
##  Median :2022-01-31     Median :2021-08-23   NA's    :12    Median :108.0
##  Mean   :2022-02-20     Mean   :2021-09-20                  Mean   :187.1
##  3rd Qu.:2023-01-04     3rd Qu.:2022-05-17                  3rd Qu.:205.0
##  Max.   :2023-04-04     Max.   :2023-02-24                  Max.   :879.0
##  NA's   :15             NA's   :19                          NA's   :19
##  lesion_burden bone_marrow_lesions subtentorial_lesions gadolinium_contrast
##  No  : 5       No  : 4             No  :11              No  :14
##  Yes :13       Yes :14             Yes : 6              Yes : 4
##  NA's:12       NA's:12             NA's:13              NA's:12
##
##
```

```
##
##
##       clinical_presentation full_recovery prognosis_at_onset  percent_th17
##  bone_marrow     : 4        No  : 7       Positive: 7        Min.   : NA
##  sensory_symptoms: 4        Yes :11       Negative: 8        1st Qu.: NA
##  optic_neuritis  : 3        NA's:12       NA's    :15        Median : NA
##  sensoriale      : 2                                         Mean   :NaN
##  brain_stem      : 1                                         3rd Qu.: NA
##  (Other)         : 4                                         Max.   : NA
##  NA's            :12                                         NA's   :30
##  treg_il10_pos treg_cd39_pos      age
##  Min.   : NA   Min.   : NA   Min.   :23.00
##  1st Qu.: NA   1st Qu.: NA   1st Qu.:33.50
##  Median : NA   Median : NA   Median :42.00
##  Mean   :NaN   Mean   :NaN   Mean   :41.53
##  3rd Qu.: NA   3rd Qu.: NA   3rd Qu.:49.25
##  Max.   : NA   Max.   : NA   Max.   :59.00
##  NA's   :30    NA's   :30
```

The cohort is composed by 30 patients affected by multiple sclerosis recruited from multiple centers (Clerico, Molinette, Napoli, CRESM). All of them were treated with cladribine (Mavenclad). The longitudinal clinical data were collected at 6-month time-points for 3 years.

The population is predominantly female, with 20 females (66.7%) and 9 males (30.0%). At baseline (T0), the mean age was 37.37 years (± 9.89), with a median of 38.50 years, ranging from 20 to 55 years, indicating a relatively young cohort. Disease activity was predominantly classified as higher activity. Disability at baseline, measured by EDSS, showed generally low values reflected by a median of 1.5 and 26.7% of patients with a EDSS 1. Regarding treatment history, 18 patients (60.0%) were naive, while 11 (36.7%) had received prior therapies. Glucocorticoid treatment before T0 was reported in 12 patients (40.0%), whereas 6 patients (20.0%) had not received such treatment and data were missing for 12 subjects.

Regarding lifestyle factors, the mean body mass index (BMI) was 23.54 ± 4.85, ranging from 16.80 to 32.10, with 9 missing values. Adherence to the Mediterranean diet, assessed by the MEDAS score (from 0 to 100), showed a mean value of 50.55 ± 15.18, with a wide range (35.00–85.70) and 8 missing observations. 7 patients (23.3%) are smokers and 17 (56.7%) are non-smokers, with 6 missing values. Physical activity is heterogeneously distributed among the cohort with 5 (16.7%) patients classified as active, 7 (23.3%) as partially active and 9 (30%) as sedentary, with 9 missing values.

```r
####tabella riassuntiva
#variabili numeriche
tab_num <- merged_data %>%
  dplyr::select(where(is.numeric)) %>%
  dplyr::select(-any_of(c("percent_th17", "treg_cd39_pos", "treg_il10_pos"))) %>%
  tidyr::pivot_longer(everything(), names_to = "variable", values_to = "val") %>%
  mutate(variable = factor(variable, levels = unique(names(merged_data)))) %>%
  dplyr::group_by(variable) %>%
  dplyr::summarise(
    All = paste0(round(mean(val, na.rm = TRUE), 2), " ± ", round(sd(val, na.rm = TRUE), 2)),
    `N missing` = as.character(sum(is.na(val))),
    .groups = "drop"
  )

#variabili categoriche
vars_cat <- merged_data %>%
  dplyr::select(where(~is.factor(.) | is.character(.) | is.ordered(.)),
                -any_of(c("ID", "NOTE", "nuova_terapia", "previous_therapy", "sample_type"))) %>%
```

```
  colnames()
lista_cat <- lapply(vars_cat, function(v) {
  miss_count <- as.character(sum(is.na(merged_data[[v]])))

levels_df <- merged_data %>%
    dplyr::group_by(level = as.character(!!sym(v))) %>%
    dplyr::summarise(n = n(), .groups = "drop") %>%
    dplyr::filter(!is.na(level)) %>%
    dplyr::mutate(
      All = paste0(n, " (", round((n / nrow(merged_data)) * 100, 1), "%)"),
      `N missing` = ""
    ) %>%
    dplyr::mutate(variable = paste0("   ", level)) %>%
    dplyr::select(variable, All, `N missing`)

  bind_rows(
    tibble(variable = v, All = "", `N missing` = miss_count),
    levels_df
  )
})

tab_cat <- bind_rows(lista_cat)

#unione
summary_table <- bind_rows(tab_num, tab_cat)

#tolgo i simboli che mandano in crash latex
summary_table$variable <- gsub("_", " ", summary_table$variable)
summary_table$All <- gsub("%", "\\\\%", summary_table$All)

knitr::kable(summary_table, caption = "Cohort Summary Table")
```

Table 1: Cohort Summary Table

| variable | All | N missing |
|---|---|---|
| MEDAS | 50.55 ± 15.18 | 8 |
| cortisone T6 | 1.29 ± 3.72 | 13 |
| cortisone T12 | 0 ± 0 | 10 |
| cortisone T18 | 0 ± 0 | 14 |
| cortisone T24 | 0.54 ± 1.33 | 17 |
| cortisone T30 | 0.58 ± 1.38 | 18 |
| cortisone T36 | 0.6 ± 1.34 | 25 |
| age at t0 | 37.37 ± 9.89 | 0 |
| bmi | 23.54 ± 4.85 | 9 |
| dna quantification | 105.36 ± 60.25 | 28 |
| days gc collection | 187.09 ± 237.89 | 19 |
| age | 41.53 ± 9.87 | 0 |
| CENTRO | | 0 |
| CLERICO | 22 (73.3%) | |
| CRESM | 4 (13.3%) | |
| MOLINETTE | 1 (3.3%) | |
| NAPOLI | 3 (10%) | |
| EDSS T0 | | 1 |

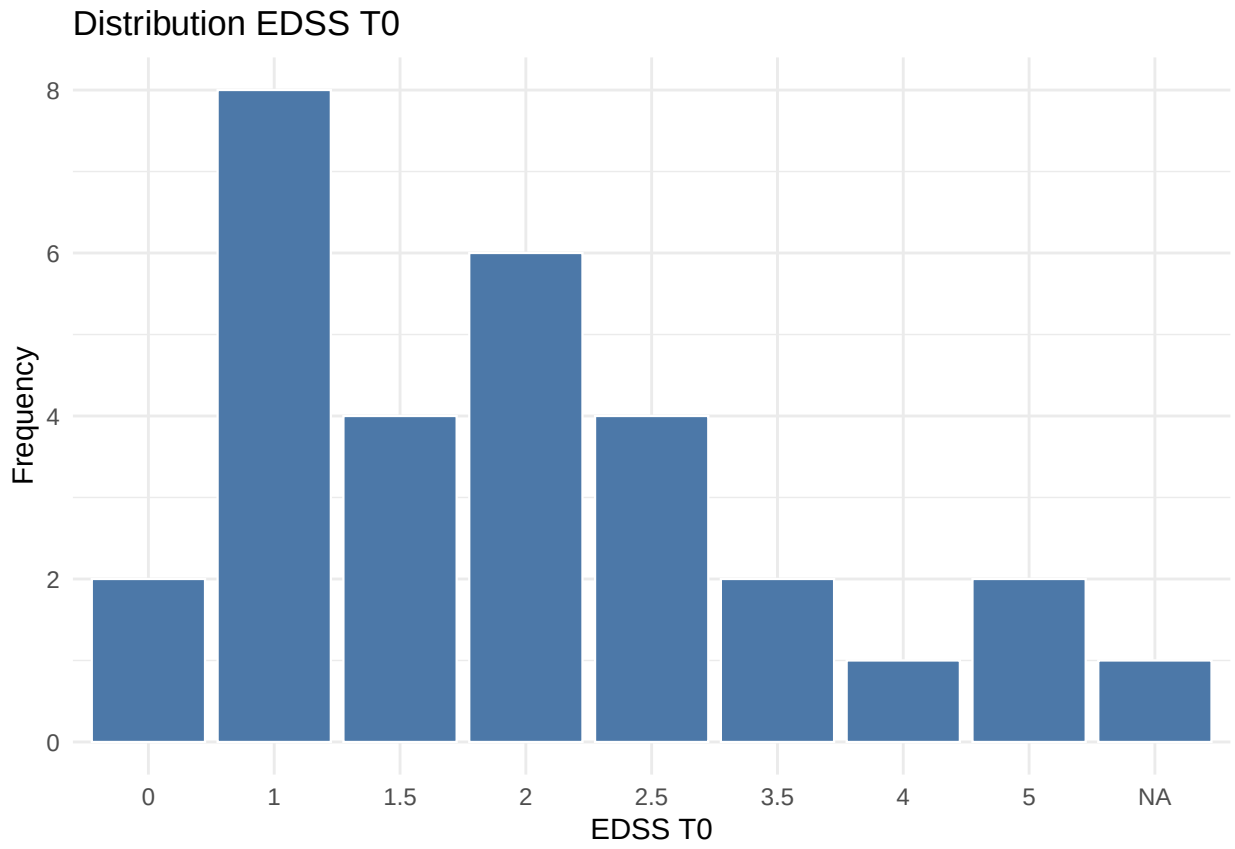
| variable | All | N missing |
|---|---|---|
| 0 | 2 (6.7%) | |
| 1 | 8 (26.7%) | |
| 1.5 | 4 (13.3%) | |
| 2 | 6 (20%) | |
| 2.5 | 4 (13.3%) | |
| 3.5 | 2 (6.7%) | |
| 4 | 1 (3.3%) | |
| 5 | 2 (6.7%) | |
| EDSS T6 | | 15 |
| 0 | 2 (6.7%) | |
| 1 | 5 (16.7%) | |
| 1.5 | 2 (6.7%) | |
| 2 | 3 (10%) | |
| 2.5 | 1 (3.3%) | |
| 3 | 1 (3.3%) | |
| 4.5 | 1 (3.3%) | |
| lesioni T6 | | 13 |
| No | 15 (50%) | |
| Yes | 2 (6.7%) | |
| relapse T6 | | 13 |
| No | 14 (46.7%) | |
| Yes | 3 (10%) | |
| EDSS T12 | | 2 |
| 0 | 5 (16.7%) | |
| 1 | 8 (26.7%) | |
| 1.5 | 4 (13.3%) | |
| 2 | 3 (10%) | |
| 2.5 | 2 (6.7%) | |
| 3 | 1 (3.3%) | |
| 3.5 | 1 (3.3%) | |
| 4 | 1 (3.3%) | |
| 4.5 | 2 (6.7%) | |
| 6.5 | 1 (3.3%) | |
| lesioni T12 | | 6 |
| No | 21 (70%) | |
| Yes | 3 (10%) | |
| relapse T12 | | 6 |
| No | 24 (80%) | |
| EDSS T18 | | 16 |
| 0 | 2 (6.7%) | |
| 1 | 4 (13.3%) | |
| 1.5 | 3 (10%) | |
| 2 | 2 (6.7%) | |
| 4 | 1 (3.3%) | |
| 4.5 | 1 (3.3%) | |
| 6.5 | 1 (3.3%) | |
| lesioni T18 | | 14 |
| No | 13 (43.3%) | |
| Yes | 3 (10%) | |
| relapse T18 | | 14 |
| No | 15 (50%) | |
| Yes | 1 (3.3%) | |

| variable | All | N missing |
|---|---|---|
| EDSS T24 | | 12 |
| 0 | 3 (10%) | |
| 1 | 4 (13.3%) | |
| 1.5 | 3 (10%) | |
| 2 | 3 (10%) | |
| 2.5 | 2 (6.7%) | |
| 4 | 2 (6.7%) | |
| 6.5 | 1 (3.3%) | |
| lesioni T24 | | 13 |
| No | 14 (46.7%) | |
| Yes | 3 (10%) | |
| relapse T24 | | 13 |
| No | 15 (50%) | |
| Yes | 2 (6.7%) | |
| EDSS T30 | | 18 |
| 0 | 3 (10%) | |
| 1 | 2 (6.7%) | |
| 1.5 | 3 (10%) | |
| 2 | 2 (6.7%) | |
| 4.5 | 1 (3.3%) | |
| 6.5 | 1 (3.3%) | |
| lesioni T30 | | 18 |
| No | 9 (30%) | |
| Yes | 3 (10%) | |
| relapse T30 | | 18 |
| No | 9 (30%) | |
| Yes | 3 (10%) | |
| EDSS T36 | | 25 |
| 0 | 1 (3.3%) | |
| 1 | 2 (6.7%) | |
| 1.5 | 1 (3.3%) | |
| 6.5 | 1 (3.3%) | |
| lesioni T36 | | 25 |
| No | 4 (13.3%) | |
| Yes | 1 (3.3%) | |
| relapse T36 | | 25 |
| No | 5 (16.7%) | |
| cambio terapia | | 9 |
| No | 14 (46.7%) | |
| Yes | 7 (23.3%) | |
| therapy | | 0 |
| cladribine | 30 (100%) | |
| naive | | 1 |
| No | 11 (36.7%) | |
| Yes | 18 (60%) | |
| sex | | 1 |
| F | 20 (66.7%) | |
| M | 9 (30%) | |
| smoking habit | | 6 |
| No | 17 (56.7%) | |
| Yes | 7 (23.3%) | |
| physical activity | | 9 |

| variable | All | N missing |
|---|---|---|
| active | 5 (16.7%) | |
| partially active | 7 (23.3%) | |
| sedentary | 9 (30%) | |
| antibiotic use | | 12 |
| No | 16 (53.3%) | |
| Yes | 2 (6.7%) | |
| sequencing batch | | 12 |
| 1 | 16 (53.3%) | |
| 3 | 2 (6.7%) | |
| spike in | | 12 |
| No | 16 (53.3%) | |
| Yes | 2 (6.7%) | |
| glucocorticoid treatment | | 12 |
| Positive | 18 (60%) | |
| simplified batch | | 12 |
| 1-3 | 18 (60%) | |
| disease activity | | 12 |
| higher activity | 18 (60%) | |
| gc treatment | | 12 |
| Negative | 6 (20%) | |
| Positive | 12 (40%) | |
| lesion burden | | 12 |
| No | 5 (16.7%) | |
| Yes | 13 (43.3%) | |
| bone marrow lesions | | 12 |
| No | 4 (13.3%) | |
| Yes | 14 (46.7%) | |
| subtentorial lesions | | 13 |
| No | 11 (36.7%) | |
| Yes | 6 (20%) | |
| gadolinium contrast | | 12 |
| No | 14 (46.7%) | |
| Yes | 4 (13.3%) | |
| clinical presentation | | 12 |
| bone marrow | 4 (13.3%) | |
| brain stem | 1 (3.3%) | |
| motor impairment | 1 (3.3%) | |
| multifocal | 1 (3.3%) | |
| norb/sensoriale | 1 (3.3%) | |
| optic neuritis | 3 (10%) | |
| sensoriale | 2 (6.7%) | |
| sensory symptoms | 4 (13.3%) | |
| tronco | 1 (3.3%) | |
| full recovery | | 12 |
| No | 7 (23.3%) | |
| Yes | 11 (36.7%) | |
| prognosis at onset | | 15 |
| Negative | 8 (26.7%) | |
| Positive | 7 (23.3%) | |

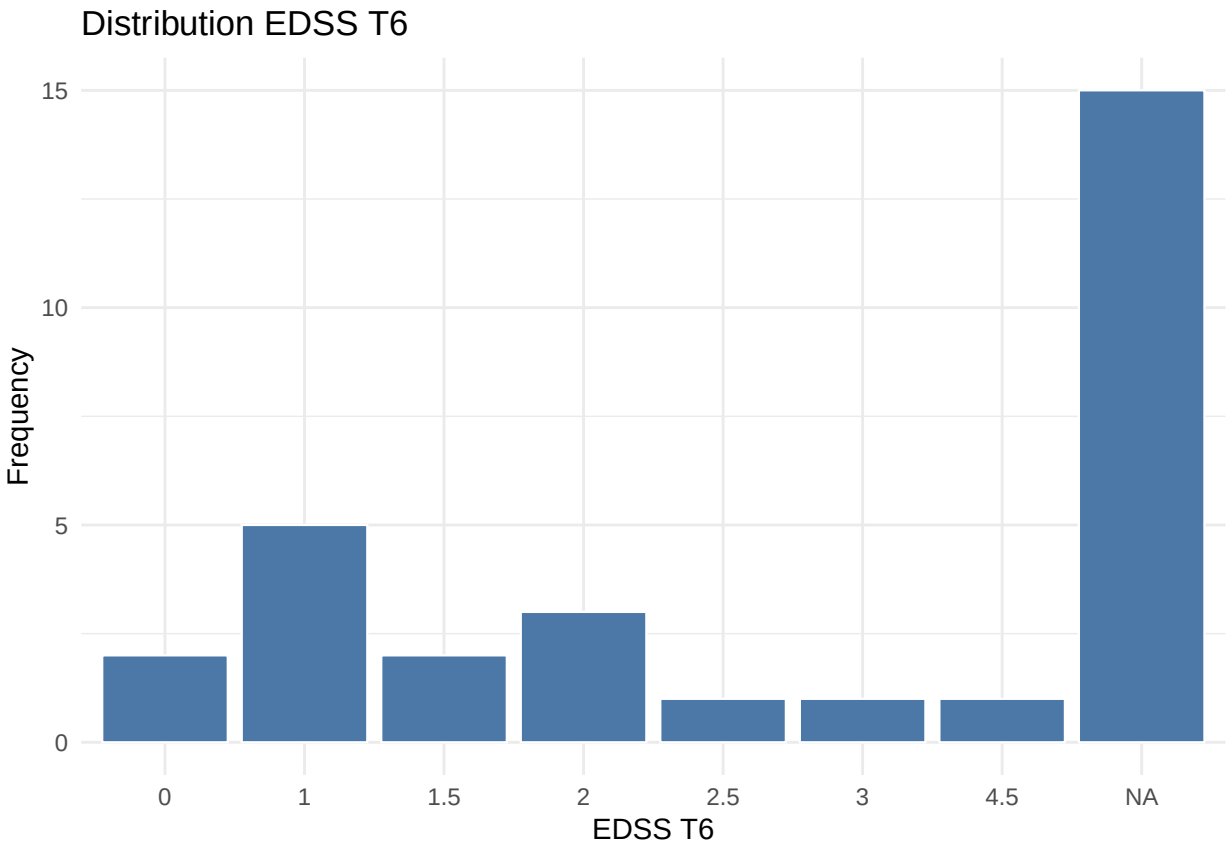
Summary graph of the distribution of the EDSS variable at time-point T0 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T0)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) + t
```



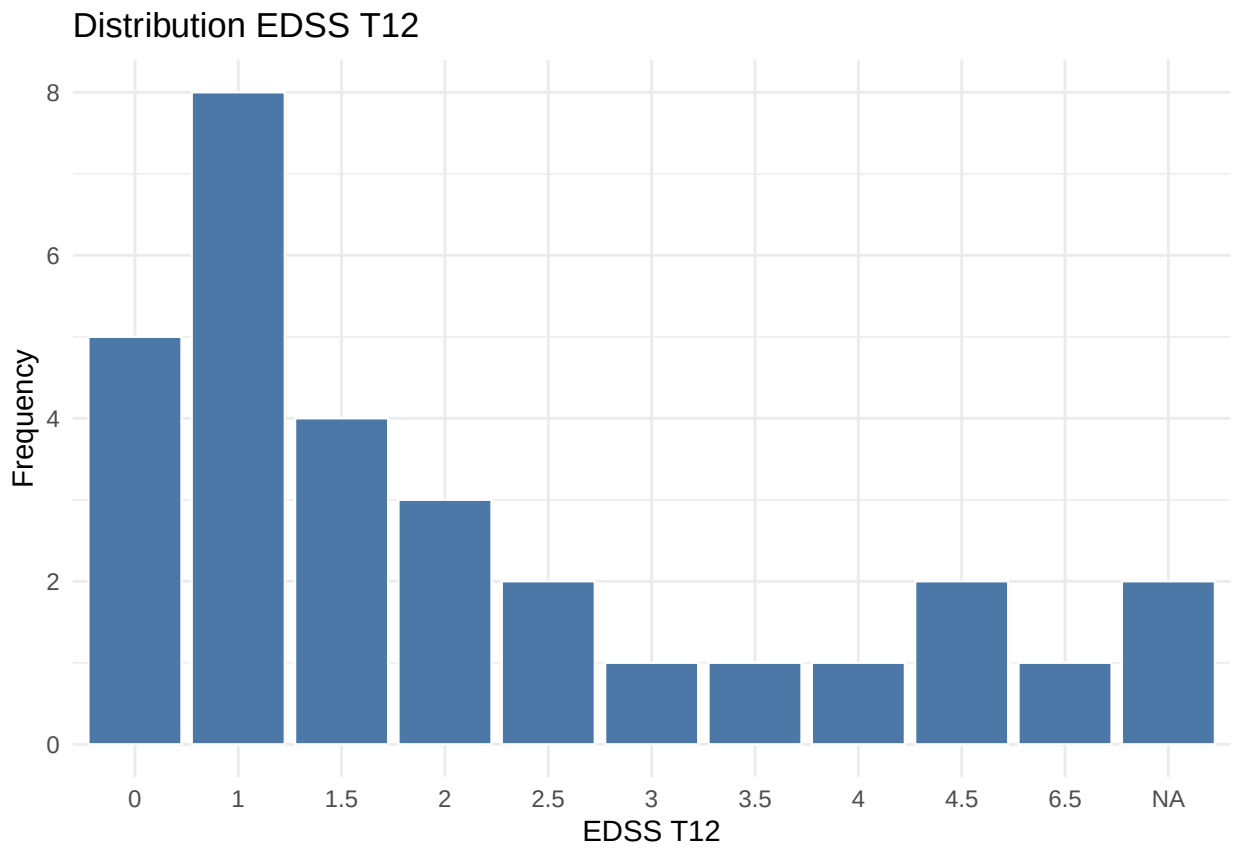

Summary graph of the distribution of the EDSS variable at time-point T6 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T6)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) + t
```



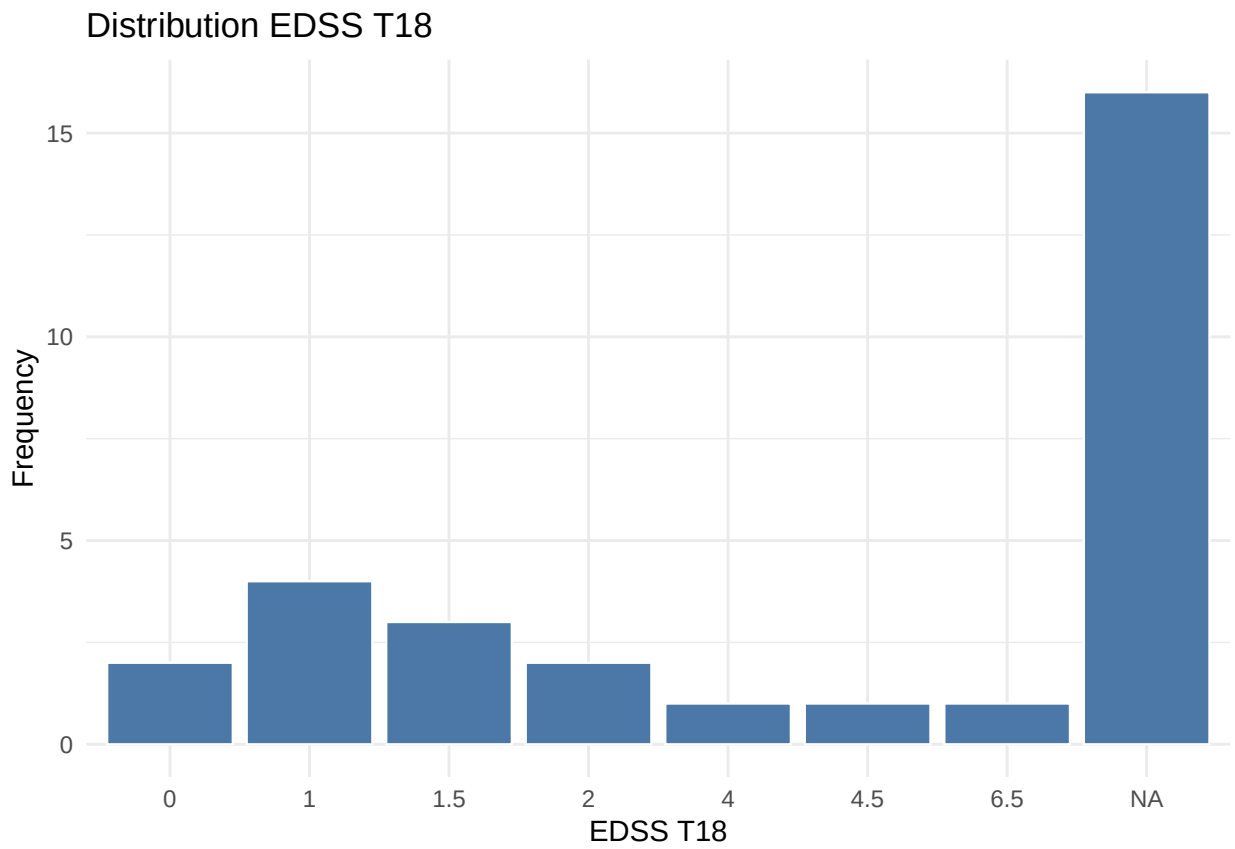

Summary graph of the distribution of the EDSS variable at time-point T12 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T12)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



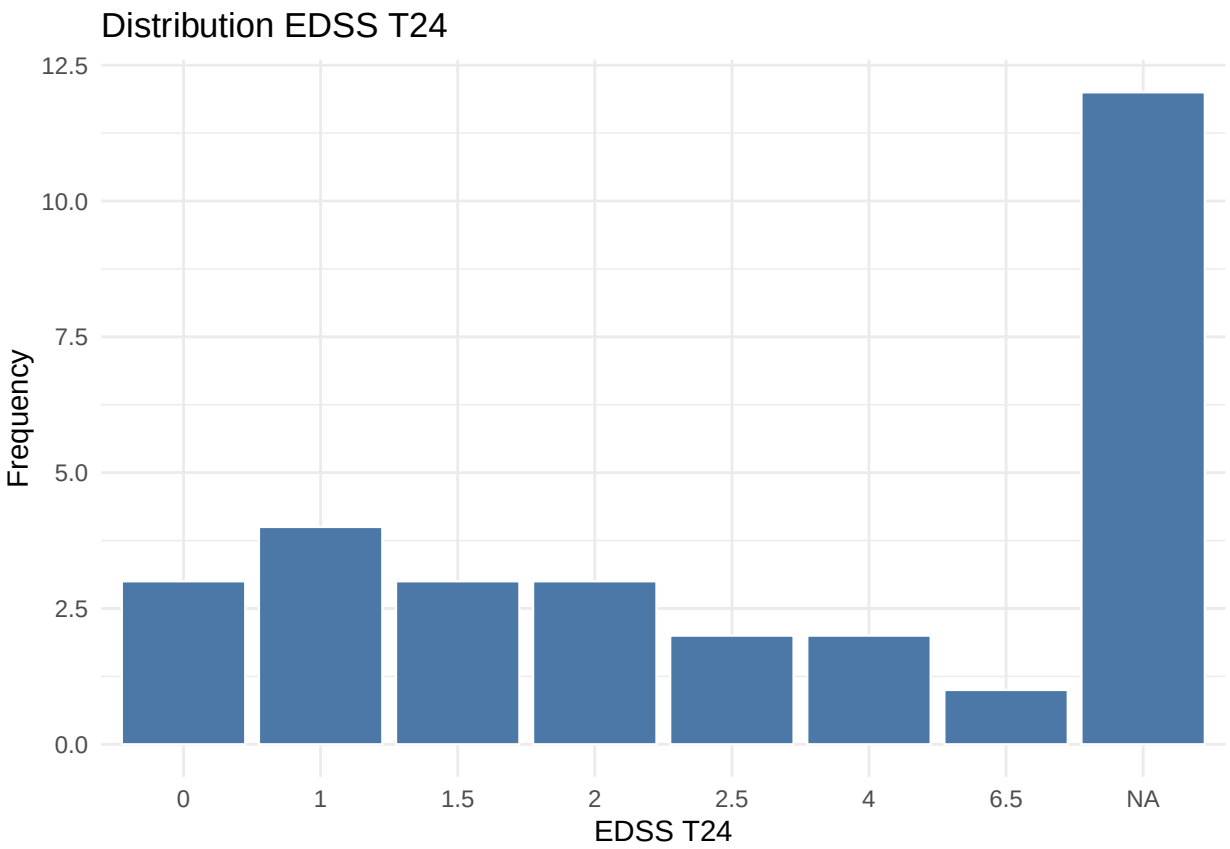

Summary graph of the distribution of the EDSS variable at time-point T18 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T18)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



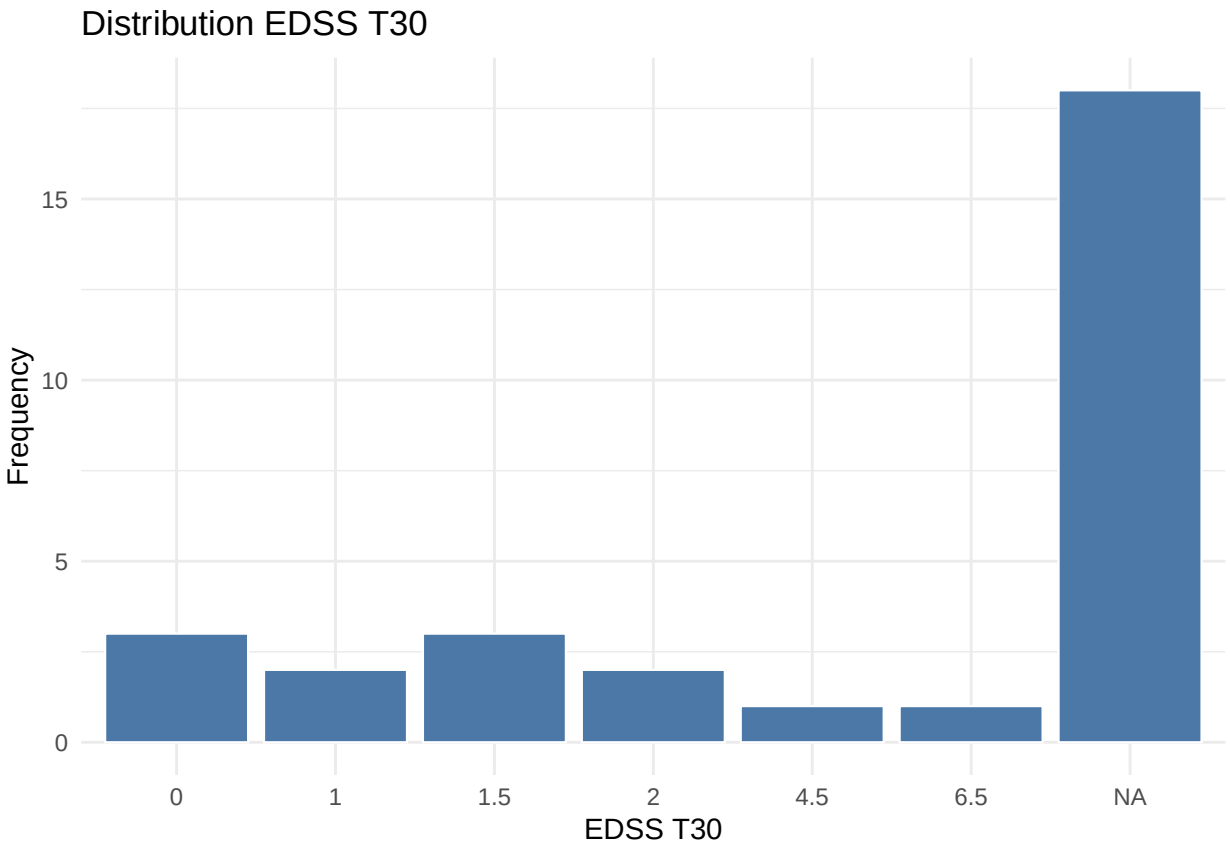

Summary graph of the distribution of the EDSS variable at time-point T24 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T24)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```



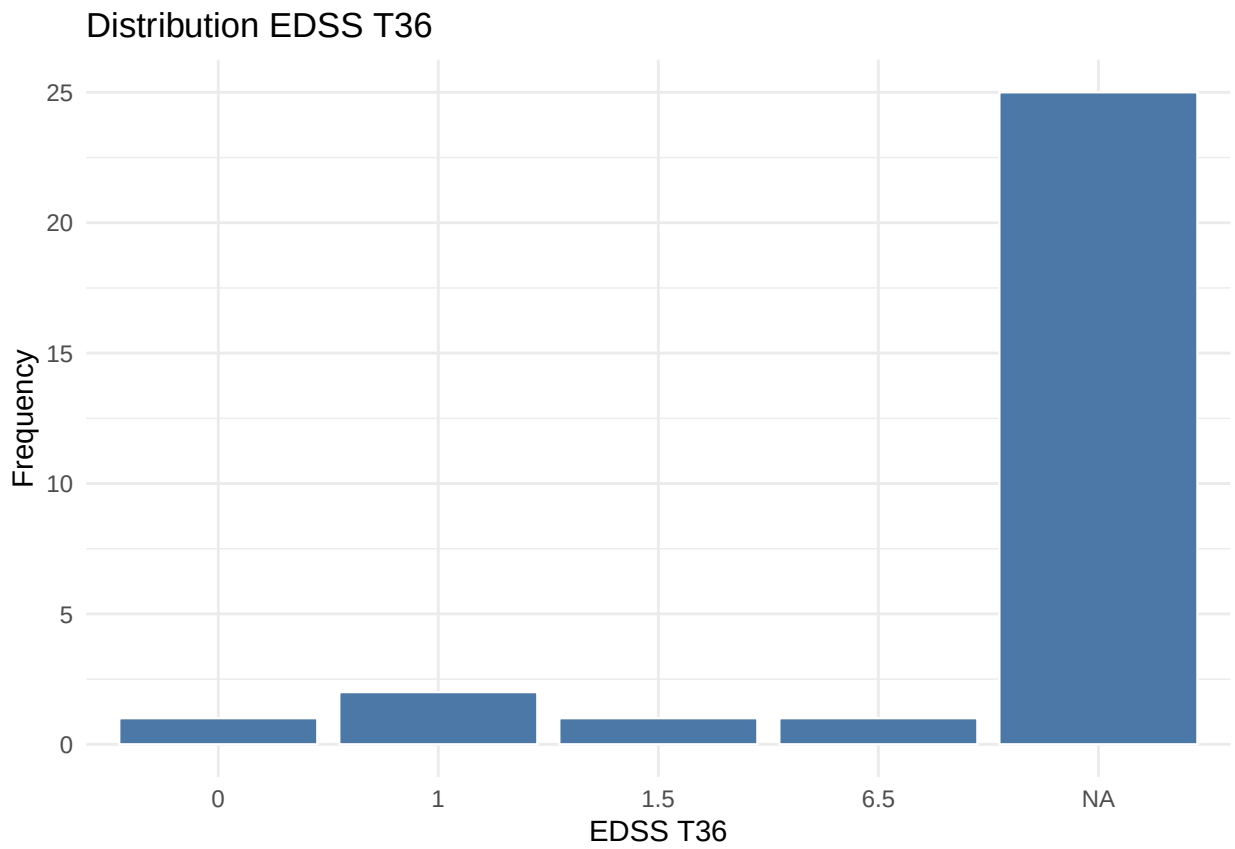

Summary graph of the distribution of the EDSS variable at time-point T30 in the cohort:

```
ggplot(merged_data, aes(x = EDSS_T30)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```




Summary graph of the distribution of the EDSS variable at time-point T36 in the cohort:

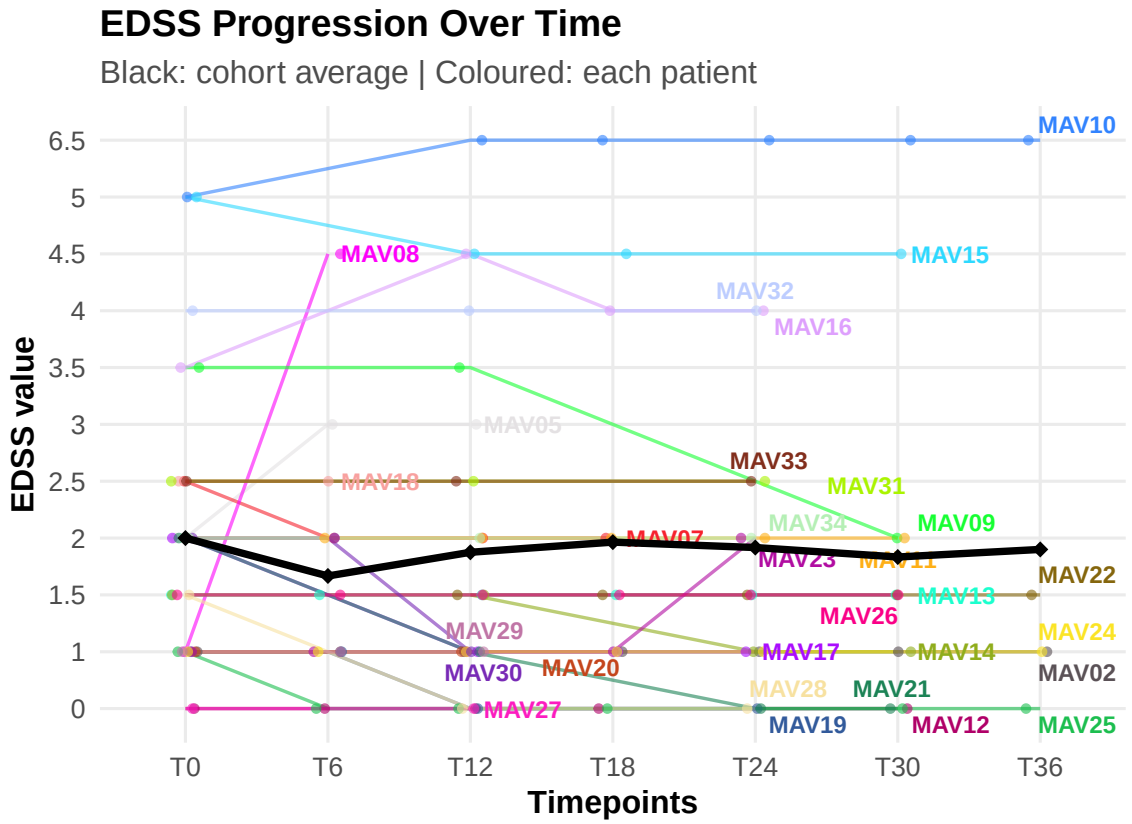
```
ggplot(merged_data, aes(x = EDSS_T36)) + geom_bar(fill = "#4C78A8", color = "white", linewidth = 0.4) +
```




Summary linepoint of the trend of the EDSS variable over time for each patient in the cohort:

```
edss_long <- merged_data %>%
  select(ID, starts_with("EDSS_T")) %>%
  pivot_longer(
    cols = starts_with("EDSS_T"),
    names_to = "timepoint",
    values_to = "EDSS"
  ) %>%
  filter(!is.na(EDSS)) %>%
  mutate(timepoint = factor(timepoint,
                            levels = paste0("EDSS_T", c(0, 6, 12, 18, 24, 30, 36)),
                            labels = paste0("T", c(0, 6, 12, 18, 24, 30, 36))))
#creo grafico
ggplot(edss_long, aes(x = timepoint, y = EDSS, group = ID)) +
  geom_line(aes(color = ID), alpha = 0.6, size = 0.6) +
  #jitter
  geom_point(aes(color = ID), alpha = 0.6, size = 1.2, position = position_jitter(width = 0.1, height =
  #aggiungo etichette ID
  geom_text_repel(data = edss_long %>%
      group_by(ID) %>%
      filter(as.numeric(timepoint) == max(as.numeric(timepoint))),
    aes(label = ID, color = ID), size = 3, fontface = "bold", nudge_x = 0.2, segment.alpha = 0, show.leg
  #Linea media
  stat_summary(aes(group = 1), fun = mean, geom = "line", size = 1.3, color = "black") +
  stat_summary(aes(group = 1), fun = mean, geom = "point", size = 2.5, color = "black", shape = 18) +
  #correzione colori per averli più distinti
```

```
  scale_color_manual(values = as.vector(pals::polychrome(length(unique(edss_long$ID)))), guide = "none")
  labs(title = "EDSS Progression Over Time",
       subtitle = "Black: cohort average | Coloured: each patient",
       x = "Timepoints", y = "EDSS value" ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank(),
        axis.title = element_text(face = "bold"),
        plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(color = "grey30"),
        plot.margin = margin(10, 40, 10, 10)) #aggiungo spazio per gli ID
```




Exploratory analysis of some relevant demographic variables: 1. EDSS and age at T0, positive and significant difference between groups only in T24 and T30 time-points (we have to consider that these two time-points have a larger number of NA).

```
# EDSS vs age_at_t0
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T0), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T0)
## S = 2958.8, p-value = 0.1547
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2712387
```

```r
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T12), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T12)
## S = 2579, p-value = 0.1286
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2942045
```

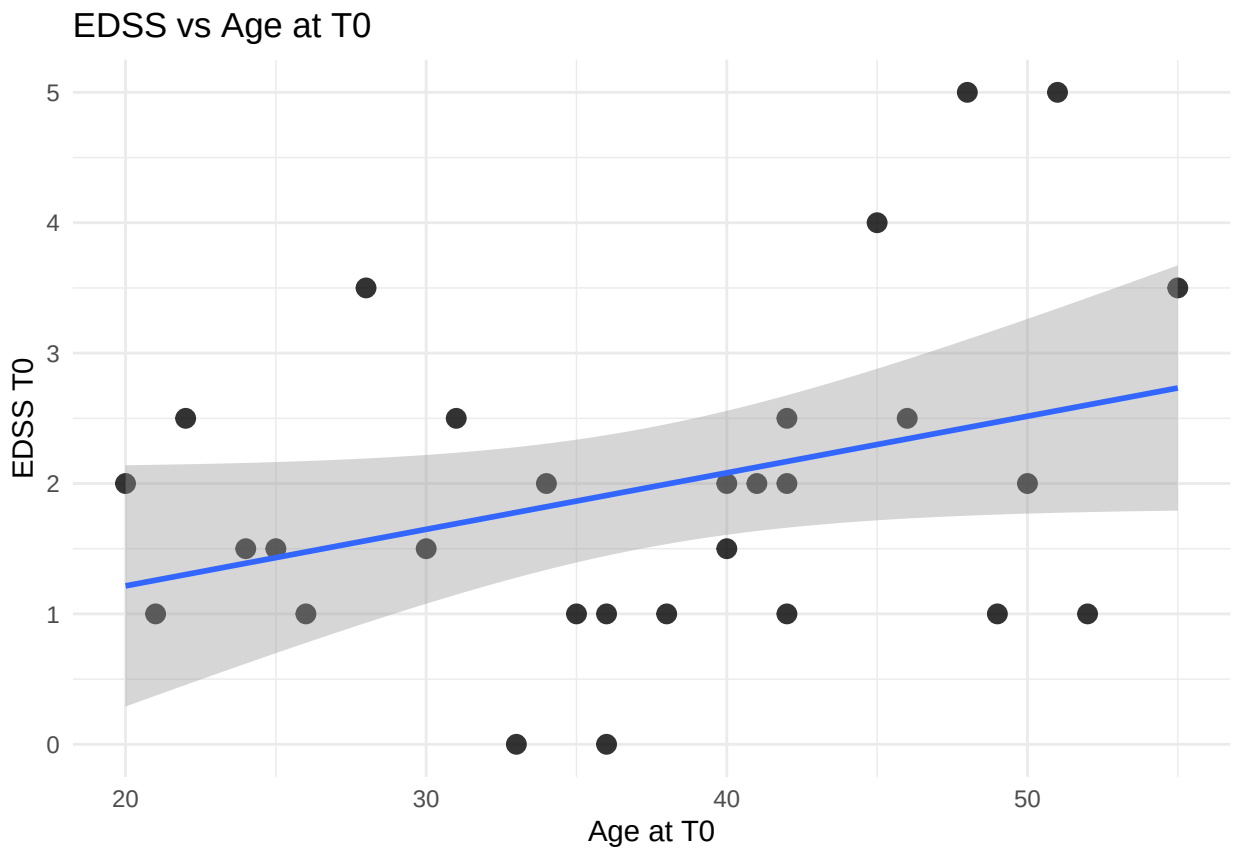
```r
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T24), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T24)
## S = 438.62, p-value = 0.01872
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.5473475
```

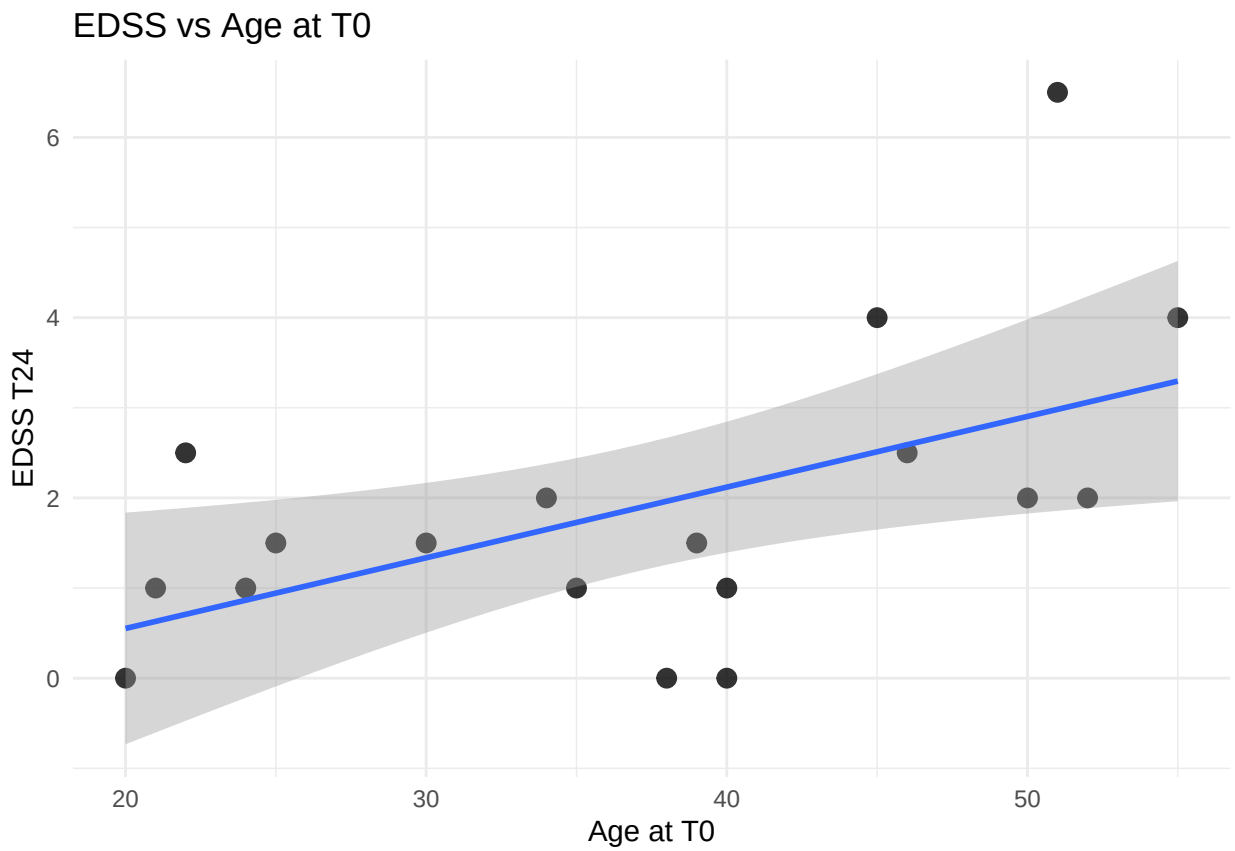
```r
cor.test(merged_data$age_at_t0, as.numeric(merged_data$EDSS_T30), method = "spearman", exact = FALSE) #
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$age_at_t0 and as.numeric(merged_data$EDSS_T30)
## S = 120.07, p-value = 0.04798
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##      rho
## 0.580163
```

```r
# linee di regressione EDSS vs age_at_t0
ggplot(merged_data, aes(x = age_at_t0, y = as.numeric(as.character(EDSS_T0)))) + geom_point(size = 3, a
```




```
ggplot(merged_data, aes(x = age_at_t0, y = as.numeric(as.character(EDSS_T24)))) + geom_point(size = 3, a
```




2. EDSS and sex, non-significant difference between groups at all time points.

```
# EDSS vs sex ##correlazione non significativa in nessun timepoint
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ sex, data = merged_data, exact = FALSE) # differenza no
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by sex
## W = 88, p-value = 0.6979
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ sex, data = merged_data, exact = FALSE) # differenza n
```

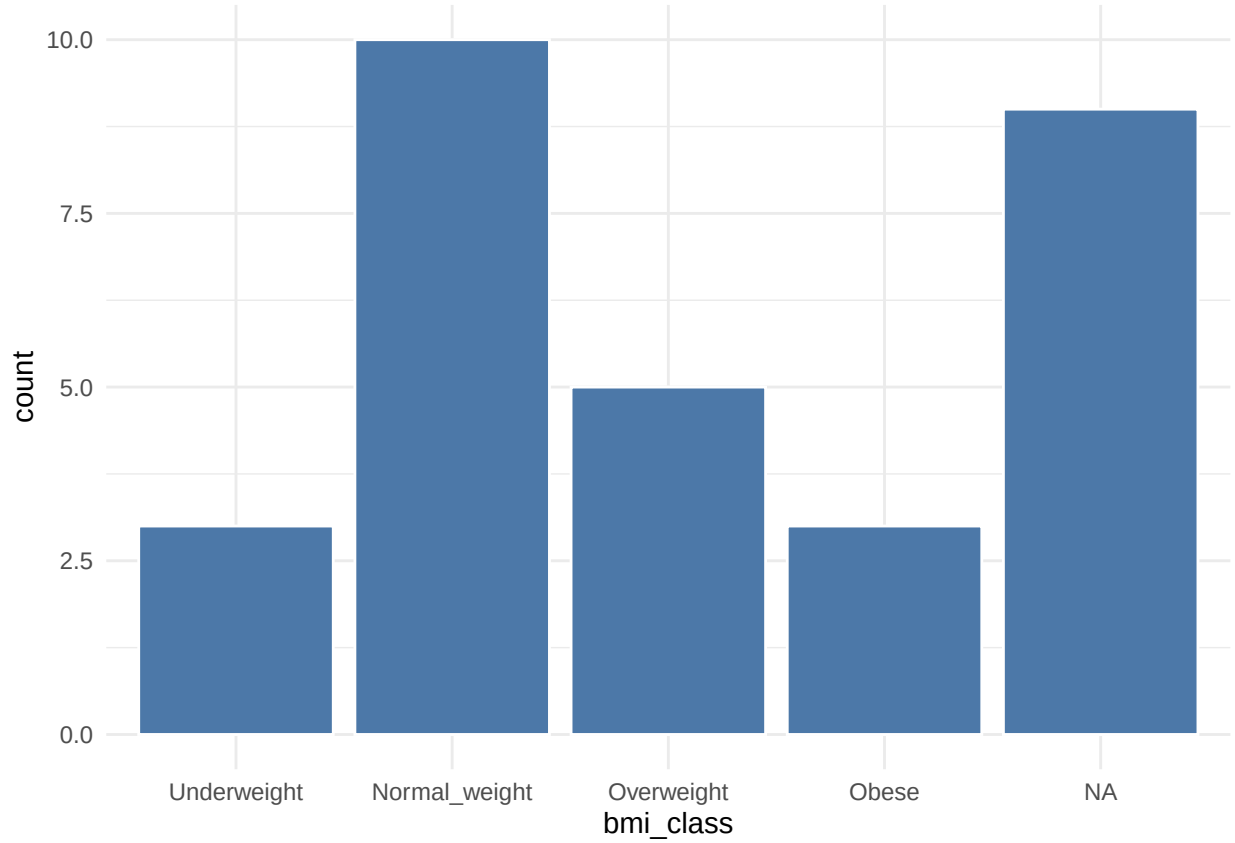
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by sex
## W = 84, p-value = 0.6851
## alternative hypothesis: true location shift is not equal to 0
```

Exploratory analysis of some relevant lifestyle factors: 1. Summary graph of the distribution of the BMI variable in the cohort:

```
merged_data$bmi_class <- factor(case_when(
    merged_data$bmi < 18.5 ~ "Underweight",
    merged_data$bmi >= 18.5 & merged_data$bmi < 25 ~ "Normal_weight",
    merged_data$bmi >= 25 & merged_data$bmi < 30 ~ "Overweight",
    merged_data$bmi >= 30 ~ "Obese",
```

```
    TRUE ~ NA_character_),
  levels = c("Underweight", "Normal_weight", "Overweight", "Obese"),ordered = TRUE)

ggplot(merged_data, aes(x = bmi_class)) + geom_bar(fill = "#4C78A8", color = "white") + theme_minimal()
```




1.1. EDSS and BMI, non-significant difference between groups at all time points.

```
# EDSS vs bmi
cor.test(merged_data$bmi, as.numeric(merged_data$EDSS_T0), method = "spearman", exact = FALSE)
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$bmi and as.numeric(merged_data$EDSS_T0)
## S = 972.79, p-value = 0.1004
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.3683168
```

```
cor.test(merged_data$bmi, as.numeric(merged_data$EDSS_T12), method = "spearman", exact = FALSE)
```

```
##
##  Spearman's rank correlation rho
##
## data:  merged_data$bmi and as.numeric(merged_data$EDSS_T12)
## S = 932.13, p-value = 0.2001
```

```
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2991508
```

2. EDSS and smoking habit, non-significant difference between groups at all time points.

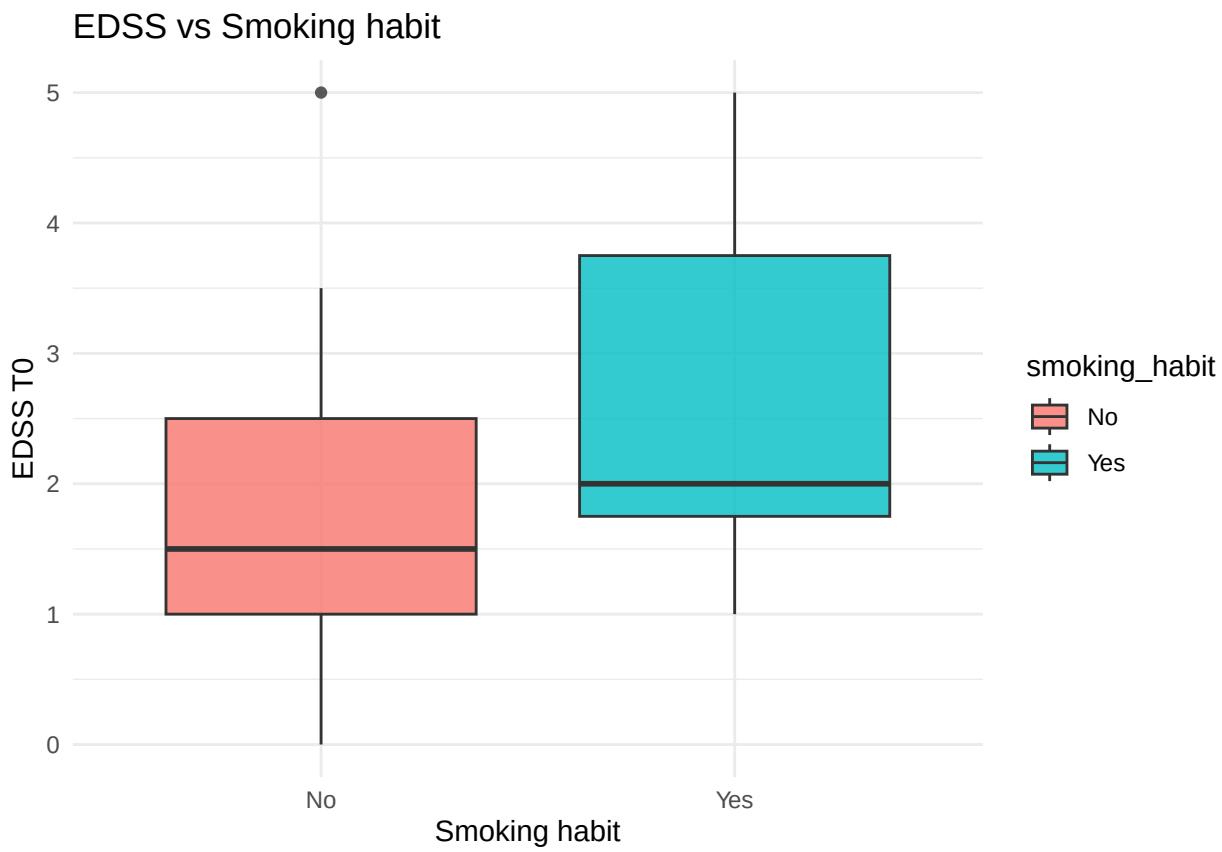
```
# EDSS vs smoking_habit #correlazione non significativa in nessun timepoint
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ smoking_habit, data = merged_data, exact = FALSE) # dif
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by smoking_habit
## W = 38.5, p-value = 0.1874
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ smoking_habit, data = merged_data, exact = FALSE) # di
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by smoking_habit
## W = 38.5, p-value = 0.2488
## alternative hypothesis: true location shift is not equal to 0
```

```
ggplot(subset(merged_data, !is.na(smoking_habit) & !is.na(EDSS_T0)), aes(x = smoking_habit, y = as.nume
```




3. EDSS and physical activity, a possible trend can be seen in the graph between the sedentary lifestyle

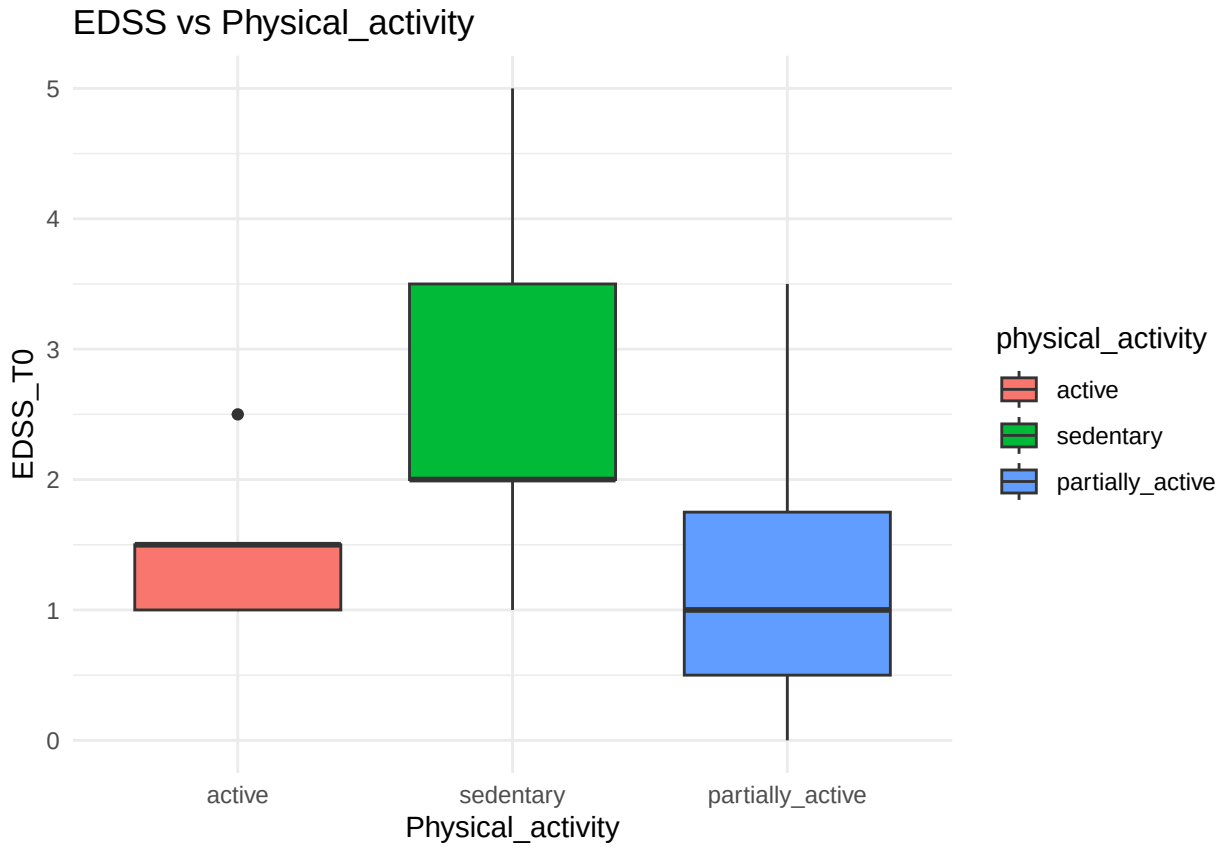
and an increase in the disability index (EDSS), but the statistical test is not significant.

```
## EDSS vs physical_activity

#T0
kruskal.test(as.numeric(as.character(EDSS_T0)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T0)) by physical_activity
## Kruskal-Wallis chi-squared = 5.168, df = 2, p-value = 0.07547
```

```
ggplot(subset(merged_data, !is.na(physical_activity) & !is.na(EDSS_T0)), aes(x = physical_activity, y =
```




```
kruskal.test(as.numeric(as.character(EDSS_T6)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T6)) by physical_activity
## Kruskal-Wallis chi-squared = 2.1859, df = 2, p-value = 0.3352
```

```
kruskal.test(as.numeric(as.character(EDSS_T12)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T12)) by physical_activity
```

```
## Kruskal-Wallis chi-squared = 4.1219, df = 2, p-value = 0.1273
```

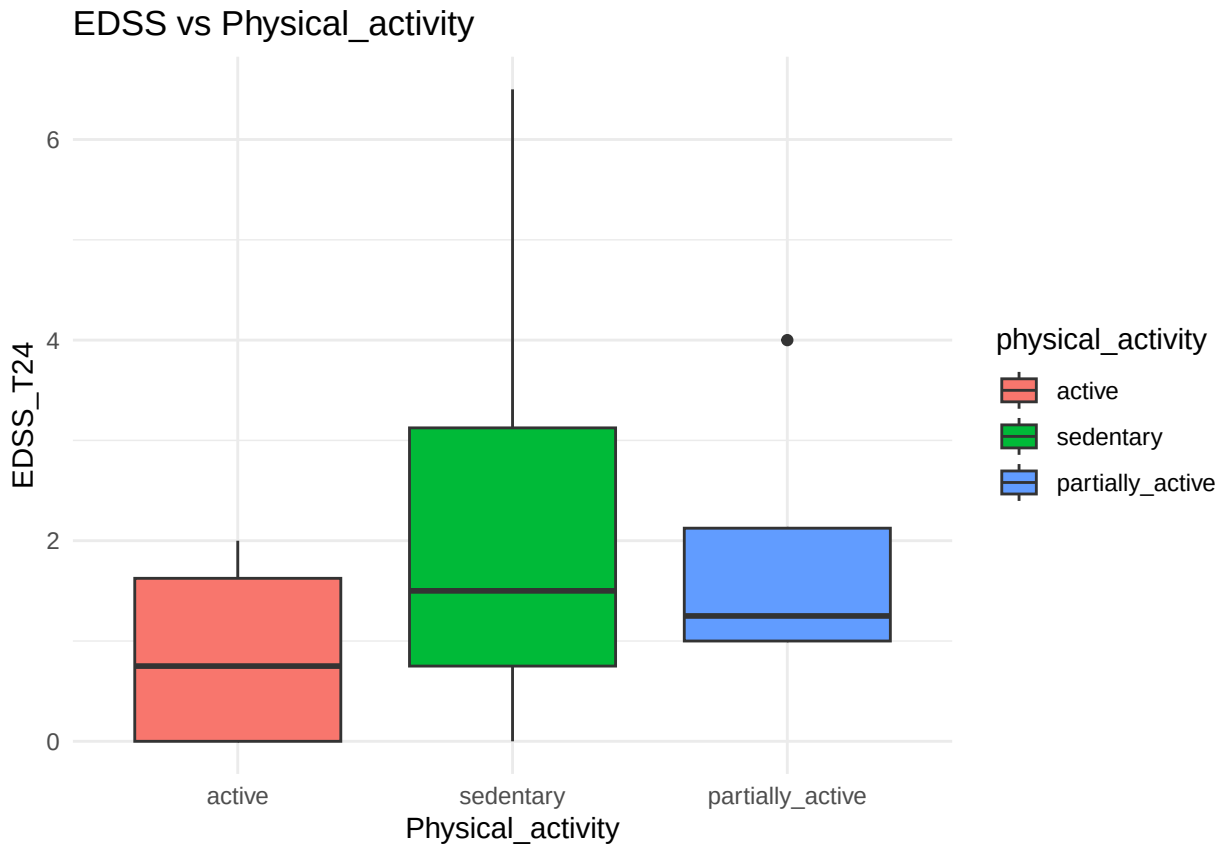
```
kruskal.test(as.numeric(as.character(EDSS_T18)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T18)) by physical_activity
## Kruskal-Wallis chi-squared = 1.5596, df = 2, p-value = 0.4585
```

```
#T24
kruskal.test(as.numeric(as.character(EDSS_T24)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T24)) by physical_activity
## Kruskal-Wallis chi-squared = 0.74728, df = 2, p-value = 0.6882
```

```
ggplot(subset(merged_data, !is.na(physical_activity) & !is.na(EDSS_T24)), aes(x = physical_activity, y =
```



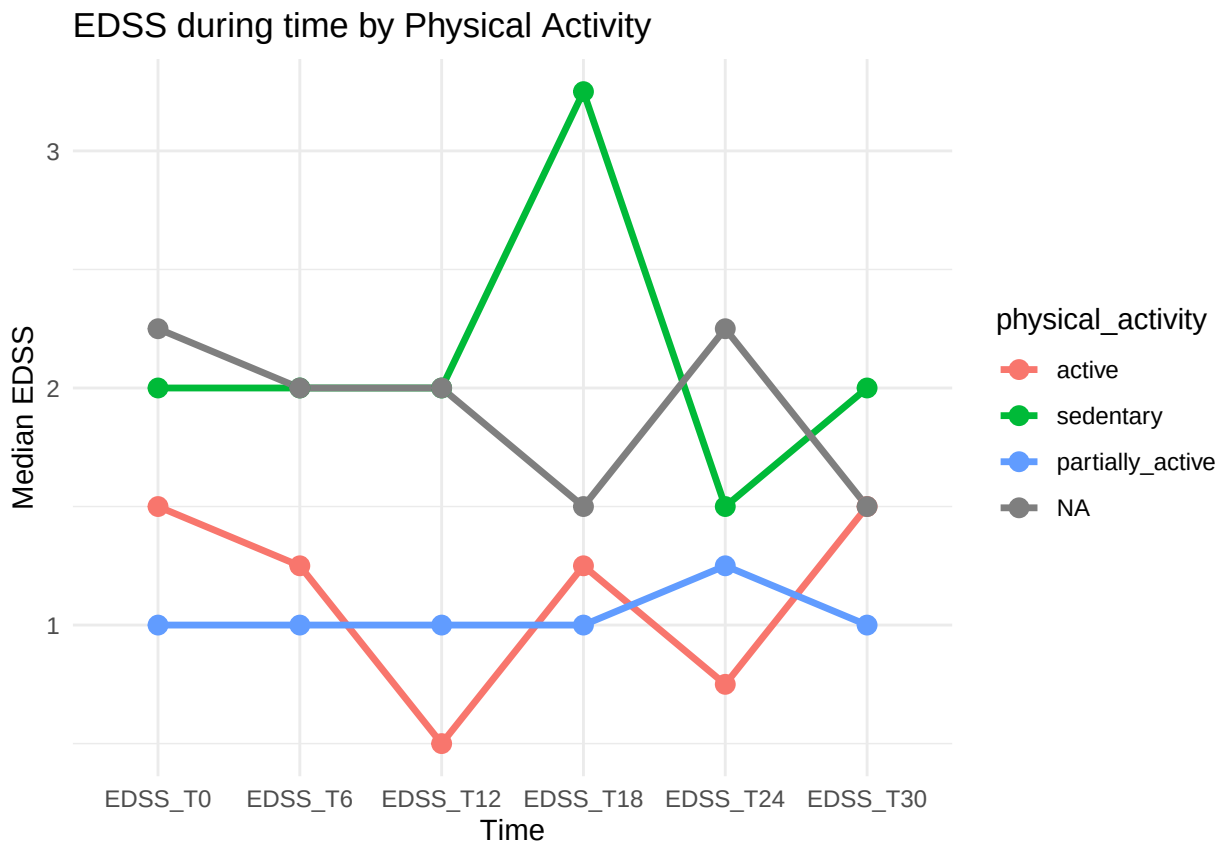

```
kruskal.test(as.numeric(as.character(EDSS_T30)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T30)) by physical_activity
## Kruskal-Wallis chi-squared = 1.0776, df = 2, p-value = 0.5835
```

```r
kruskal.test(as.numeric(as.character(EDSS_T36)) ~ physical_activity, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T36)) by physical_activity
## Kruskal-Wallis chi-squared = 0, df = 1, p-value = 1
```

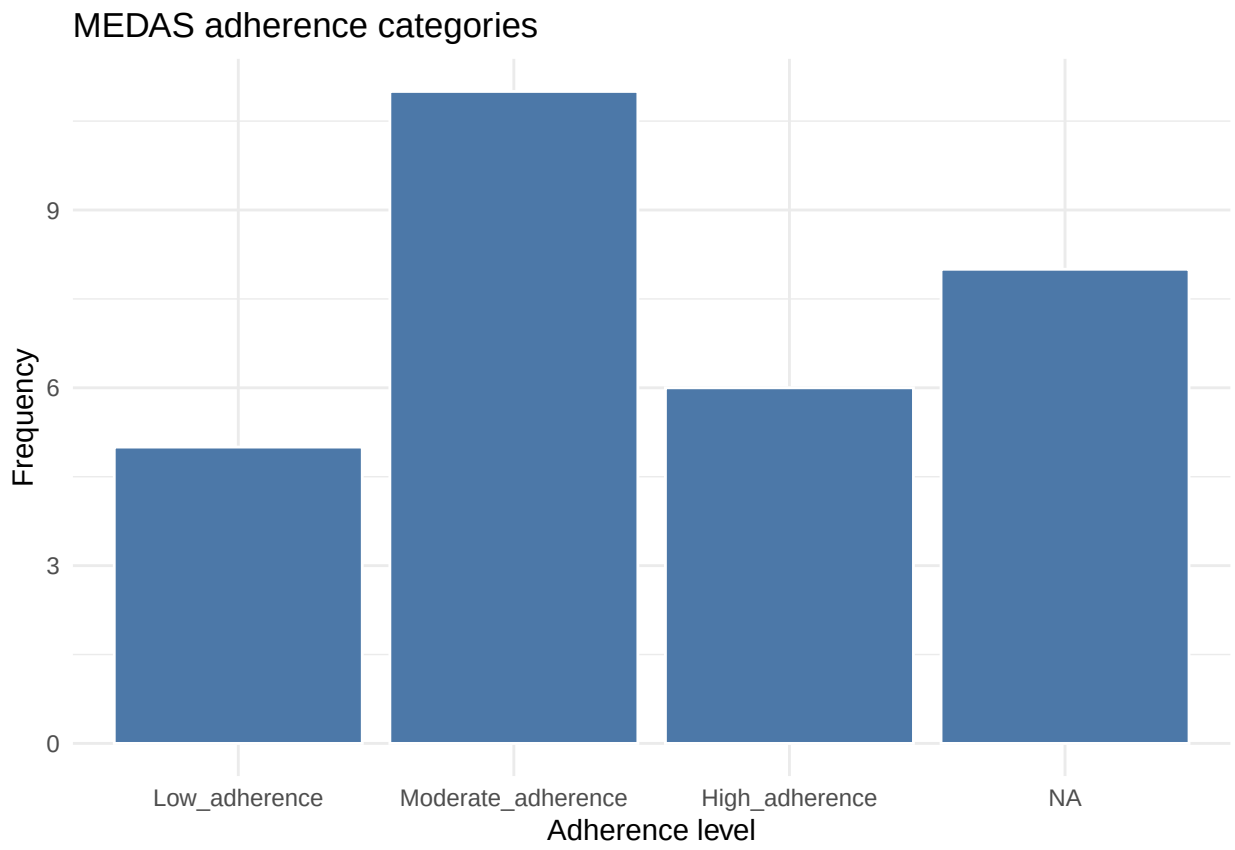
```r
#andamento totale
edss_long <- merged_data %>%
  select(ID, smoking_habit, sex, naive, physical_activity, EDSS_T0, EDSS_T6, EDSS_T12, EDSS_T18, EDSS_T
  pivot_longer(cols = starts_with("EDSS"), names_to = "time", values_to = "EDSS_val") %>%
  mutate(EDSS_val = as.numeric(as.character(EDSS_val)), sex = trimws(as.character(sex)), time = factor(
  filter(!is.na(EDSS_val))
ggplot(edss_long, aes(x = time, y = EDSS_val, color = physical_activity, group = physical_activity)) +
  stat_summary(fun = median, geom = "line", linewidth = 1.2, na.rm = TRUE) +
  stat_summary(fun = median, geom = "point", size = 3, na.rm = TRUE) +
  theme_minimal() + labs(title="EDSS during time by Physical Activity", x = "Time", y = "Median EDSS")
```




4. Summary graph of the distribution of the MEDAS variable in the cohort:

```r
merged_data$MEDAS_class <- factor(case_when(
  merged_data$MEDAS < 40 ~ "Low_adherence",
  merged_data$MEDAS >= 40 & merged_data$MEDAS < 60 ~ "Moderate_adherence",
  merged_data$MEDAS >= 60 ~ "High_adherence",
  TRUE ~ NA_character_),
  levels = c("Low_adherence", "Moderate_adherence", "High_adherence"),ordered = TRUE)
```

```
ggplot(merged_data, aes(x = MEDAS_class)) + geom_bar(fill = "#4C78A8", color = "white") + theme_minimal
```




4.1. EDSS and MEDAS, non-significant difference between groups at all time points.

```
# EDSS vs MEDAS
kruskal.test(as.numeric(as.character(EDSS_T0)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T0)) by MEDAS
## Kruskal-Wallis chi-squared = 6.0573, df = 7, p-value = 0.5331
```

```
kruskal.test(as.numeric(as.character(EDSS_T12)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T12)) by MEDAS
## Kruskal-Wallis chi-squared = 6.3061, df = 8, p-value = 0.613
```

```
kruskal.test(as.numeric(as.character(EDSS_T24)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T24)) by MEDAS
## Kruskal-Wallis chi-squared = 4.3342, df = 6, p-value = 0.6315
```

```r
kruskal.test(as.numeric(as.character(EDSS_T36)) ~ MEDAS, data = merged_data)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  as.numeric(as.character(EDSS_T36)) by MEDAS
## Kruskal-Wallis chi-squared = 3, df = 3, p-value = 0.3916
```

Exploratory analysis of some relevant clinical-radiological variables 1. Age at T0 and relapse, non-significant difference between groups at all time points.

```r
# age_at_t0 vs relapse
t.test(age_at_t0 ~ relapse_T6, data = merged_data)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by relapse_T6
## t = -0.10904, df = 5.3868, p-value = 0.9171
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -10.890771   9.986009
## sample estimates:
##  mean in group No mean in group Yes
##          35.21429          35.66667
```

```r
t.test(age_at_t0 ~ relapse_T24, data = merged_data)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by relapse_T24
## t = -0.48521, df = 1.1154, p-value = 0.7056
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -134.5081  122.0415
## sample estimates:
##  mean in group No mean in group Yes
##          36.26667          42.50000
```

2. Age at T0 and lesions, positive and significant difference between groups on T6 an T12.

```r
# age_at_t0 vs lesions
t.test(age_at_t0 ~ lesioni_T6, data = merged_data) # correlazione SIGNIFICATIVA (di poco)
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T6
## t = 2.1402, df = 14.815, p-value = 0.0494
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##   0.01636283 10.85030384
## sample estimates:
##  mean in group No mean in group Yes
##          35.93333          30.50000
```

```r
t.test(age_at_t0 ~ lesioni_T12, data = merged_data) # correlazione SIGNIFICATIVA
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T12
## t = -3.616, df = 4.8659, p-value = 0.01602
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -20.356548  -3.357738
## sample estimates:
##  mean in group No mean in group Yes
##          34.47619          46.33333
```

```r
t.test(age_at_t0 ~ lesioni_T24, data = merged_data) # correlazione non significativa
```

```
##
##  Welch Two Sample t-test
##
## data:  age_at_t0 by lesioni_T24
## t = -1.2185, df = 2.6242, p-value = 0.3212
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
##  -38.82454  18.58644
## sample estimates:
##  mean in group No mean in group Yes
##          35.21429          45.33333
```

3. EDSS and lesion burden, non-significant difference between groups at all time points, but looking at the graph you can notice a possible trend of association of higher EDSS values to patients with lesions.

```r
#lesion burden
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ lesion_burden, data = merged_data, exact = FALSE)
```

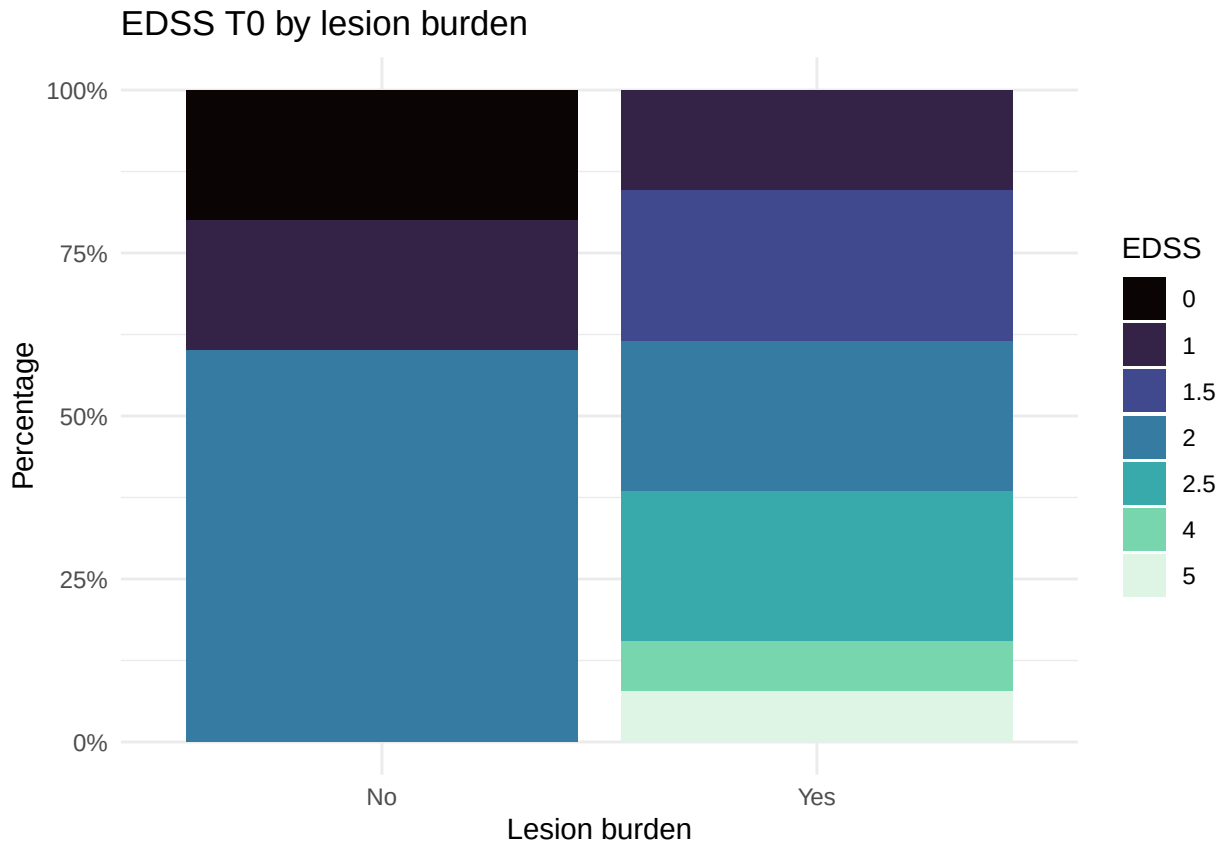
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by lesion_burden
## W = 20.5, p-value = 0.2452
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ lesion_burden, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by lesion_burden
## W = 21.5, p-value = 0.3911
## alternative hypothesis: true location shift is not equal to 0
```

```r
merged_data %>%
  filter(!is.na(EDSS_T0), !is.na(lesion_burden)) %>%
  mutate(
    EDSS_T0 = as.factor(EDSS_T0),
    lesion_burden = factor(lesion_burden, labels = c("No", "Yes"))) %>%
  ggplot(aes(x = lesion_burden, fill = EDSS_T0)) +
  geom_bar(position = "fill") + scale_y_continuous(labels = scales::percent) +
```

```
  scale_fill_viridis_d(option = "mako") + theme_minimal() + labs(title = "EDSS T0 by lesion burden", x =
```




4. EDSS and bone marrow lesions, the wilcox test shows a significant difference only at the time-point T12. All other time-points do not have a significant difference, but looking at the graph you can notice a trend of association of higher EDSS values to patients with lesions.

```
#Bone marrow lesions
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ bone_marrow_lesions, data = merged_data, exact = FALSE)
```

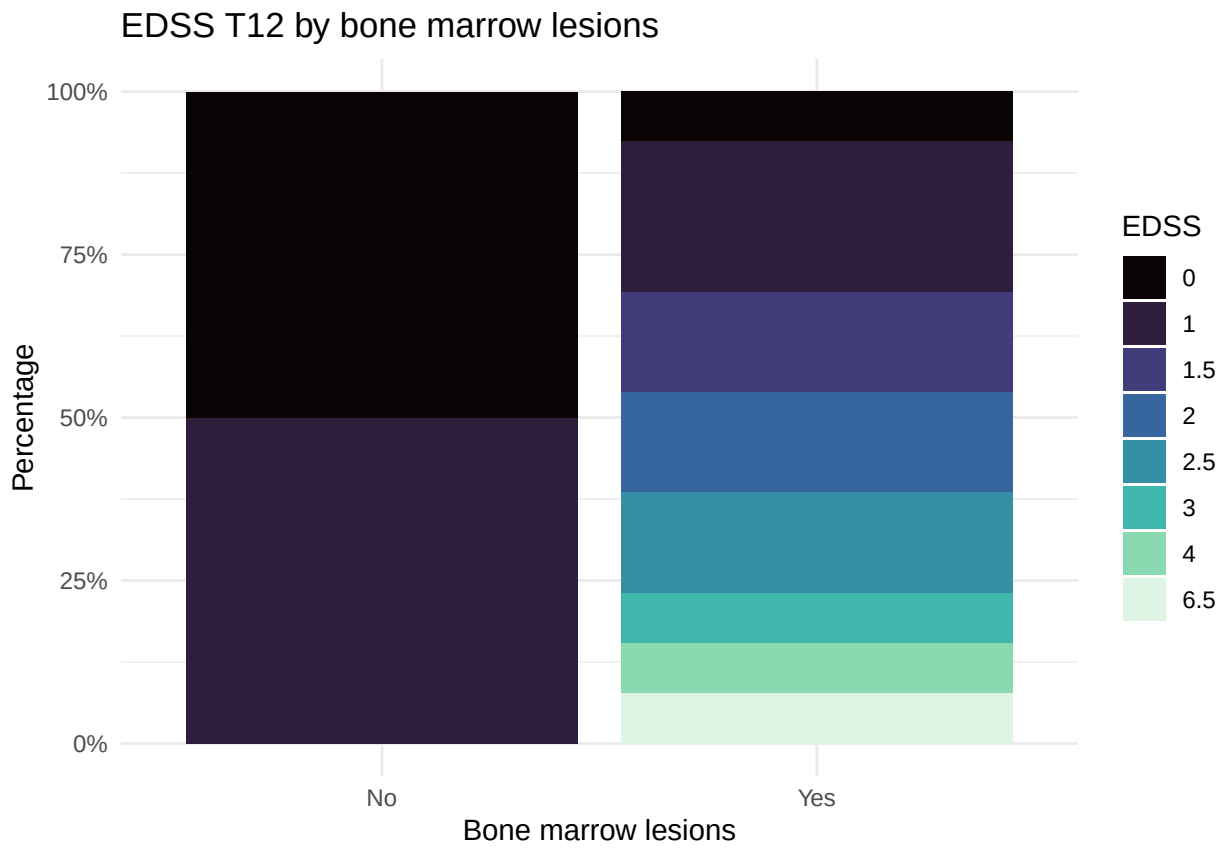
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by bone_marrow_lesions
## W = 20, p-value = 0.4142
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ bone_marrow_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by bone_marrow_lesions
## W = 6, p-value = 0.02474
## alternative hypothesis: true location shift is not equal to 0
```

```
merged_data %>%
  filter(!is.na(EDSS_T12), !is.na(bone_marrow_lesions)) %>%
  mutate(
```

```
    EDSS_T12 = as.factor(EDSS_T12),
    bone_marrow_lesions = factor(bone_marrow_lesions, labels = c("No", "Yes"))) %>%
  ggplot(aes(x = bone_marrow_lesions, fill = EDSS_T12)) +
  geom_bar(position = "fill") + scale_y_continuous(labels = scales::percent) +
  scale_fill_viridis_d(option = "mako") + theme_minimal() + labs(title = "EDSS T12 by bone marrow lesion
```




5. EDSS and subtentorial lesions, non-significant difference between groups at all time points.

```
#subtentorial lesions
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ subtentorial_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by subtentorial_lesions
## W = 32, p-value = 0.9591
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ subtentorial_lesions, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by subtentorial_lesions
## W = 24, p-value = 0.7289
## alternative hypothesis: true location shift is not equal to 0
```

6. EDSS and gadolinium contrast, non-significant difference between groups at all time points.

```r
#gadolinium contrast
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ gadolinium_contrast, data = merged_data, exact = FALSE)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by gadolinium_contrast
## W = 31.5, p-value = 0.744
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(as.numeric(as.character(EDSS_T12)) ~ gadolinium_contrast, data = merged_data, exact = FALSE)
```

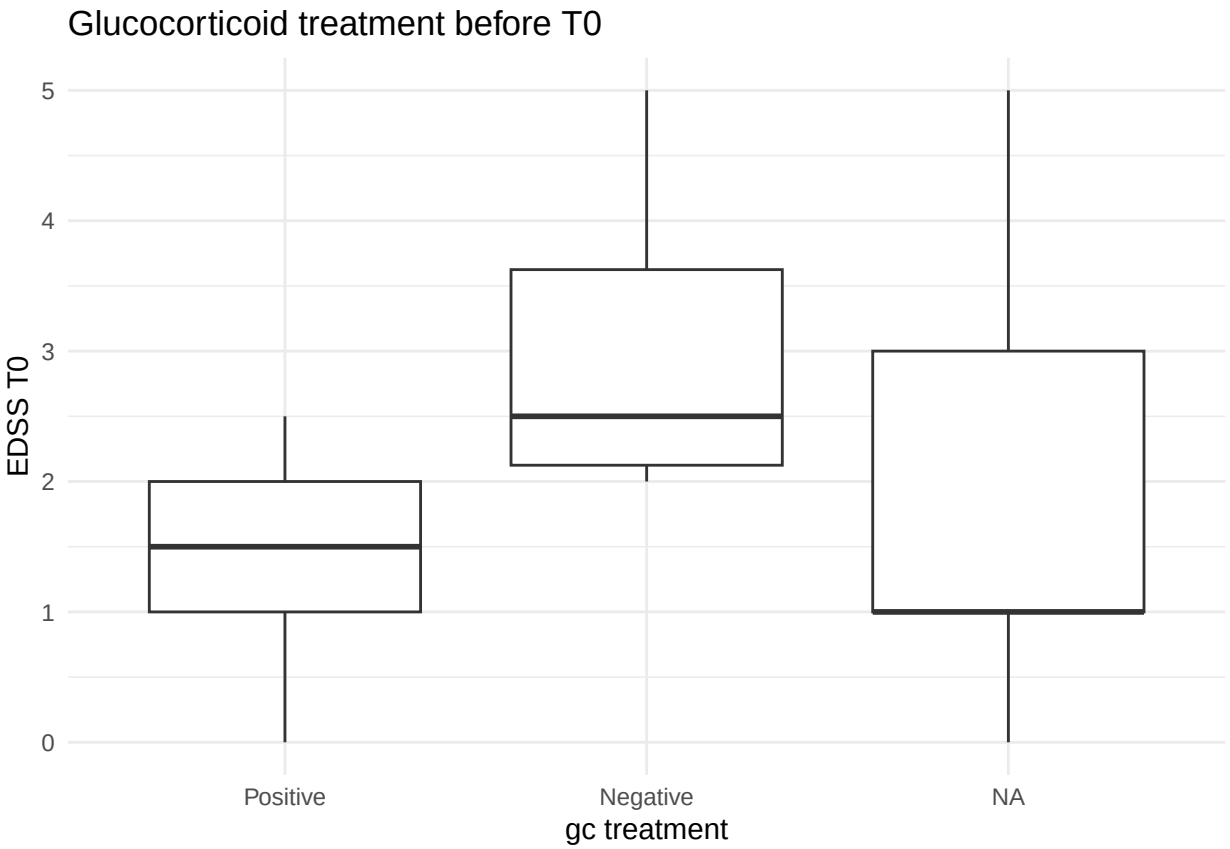
```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T12)) by gadolinium_contrast
## W = 25, p-value = 0.6538
## alternative hypothesis: true location shift is not equal to 0
```

7. Glucocorticoid treatment before baseline and EDSS, significant positive difference between groups in the T0 timepoint.

```r
#glucocorticoid treatment before T0 e edss T0 #SIGNIFICATIVO
wilcox.test(as.numeric(as.character(EDSS_T0)) ~ gc_treatment, data = merged_data)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  as.numeric(as.character(EDSS_T0)) by gc_treatment
## W = 7, p-value = 0.00621
## alternative hypothesis: true location shift is not equal to 0
```

```r
ggplot(merged_data, aes(x = gc_treatment, y = as.numeric(as.character(EDSS_T0)))) + geom_boxplot() + th
```




Overall, no statistically significant associations were observed between EDSS and most clinical or lifestyle variables across time points. Notably, a significant positive association was observed between EDSS and bone marrow lesions, specifically at the T12 time-point. Additionally, glucocorticoid treatment showed a significant positive difference between groups at T0. Regarding age, while age at baseline was generally associated with higher EDSS values at later stages, a specific positive and significant difference was found between age at T0 and the presence of lesions at both T6 and T12. Physical activity showed a potential trend, with more sedentary patients displaying a higher EDSS values over time, but the statistical tests are not significant. Although these results remain robust, it is important to consider that the findings at later time points may be partially influenced by missing data.